

Female Reproductive System

Integrated Cases between

Histology Cell Biology & Biochemistry

Section A:

Research Article: Self-directed learning. Ideal Nutrition in pregnancy and lactation.

Use library resources or open resources (eg. Google Scholar) to find a peer reviewed article on Ideal Nutrition in pregnancy and lactation. Use the following objectives as a guide to review the selected paper. Once you can answer most of these questions, then you have completed your review and are fully prepared for discussion.

- I. For the better health of the fetus and the mother, what are the ideal nutritional recommendations that as a doctor you will give?
- II. During the ultrasound examination of a pregnant patient at 6 months, the physician notices a large translucent swelling at the lumbosacral region of the fetus.

Based on your clinical knowledge, what is the congenital anomaly present in this patient and how could this anomaly have been prevented nutritionally?

Section B: Integrated Case

A 41-year-old woman comes to the emergency department with a 1-week history of light intermittent vaginal bleeding.

She says that the bleeding has not soaked through her pad. She had her first period (menarche) at age 13 and has had irregular periods ever since with cycles that last 35 to 50 days, and occasional heavy bleeding. Her last menstrual period was 53 days ago. She is currently sexually active. She is not on any form of contraception. She has no family history of bleeding disorders, cancer, or heart disease.

Her blood pressure is 100/60 mmHg, pulse is 100/min, respirations are 20/min and temperature is 37°C (98.6°F).

Physical examination shows conjunctival pallor and breast engorgement with darkening of the nipples.

Blood is collected for analysis and ultrasound is performed.

Questions

1. Is it possible that she could be menopausal, or is she pregnant? How would you confirm if this patient is pregnant?
2. Based on her last menstrual period, which week of pregnancy is she in?
3. What are the physiological and anatomical changes noticed both on maternal and fetal side at the early stages of pregnancy?
4. Which of the hormones are predominant in the early part of pregnancy, and what supports the corpus luteum?
5. What is the function of prolactin in pregnancy?
6. Based on her age, she wants to undergo genetic screening and other assays to rule out any form of birth defects. What are the common techniques used for this process?

Under the current recommendations, pregnant women are offered prenatal screening during the first and second trimesters of pregnancy.

I. What are the different types of non-invasive and invasive screening tests? Mention their optimal timing.

Non-Invasive	Invasive
a.	
b.	

II. Distinguish between first and second trimester maternal serum screening tests. Mention the findings of maternal serum screening tests found in each of the following diseases (in addition to nuchal translucency):

- (a) Trisomy 21
- (b) Trisomy 18
- (c) Trisomy 13
- (d) Neural Tube Defects

III. She is offered NIPS by next-generation sequencing of cell-free fetal DNA. Describe this test, and explain how it can identify a fetus that is at risk of having Down syndrome.

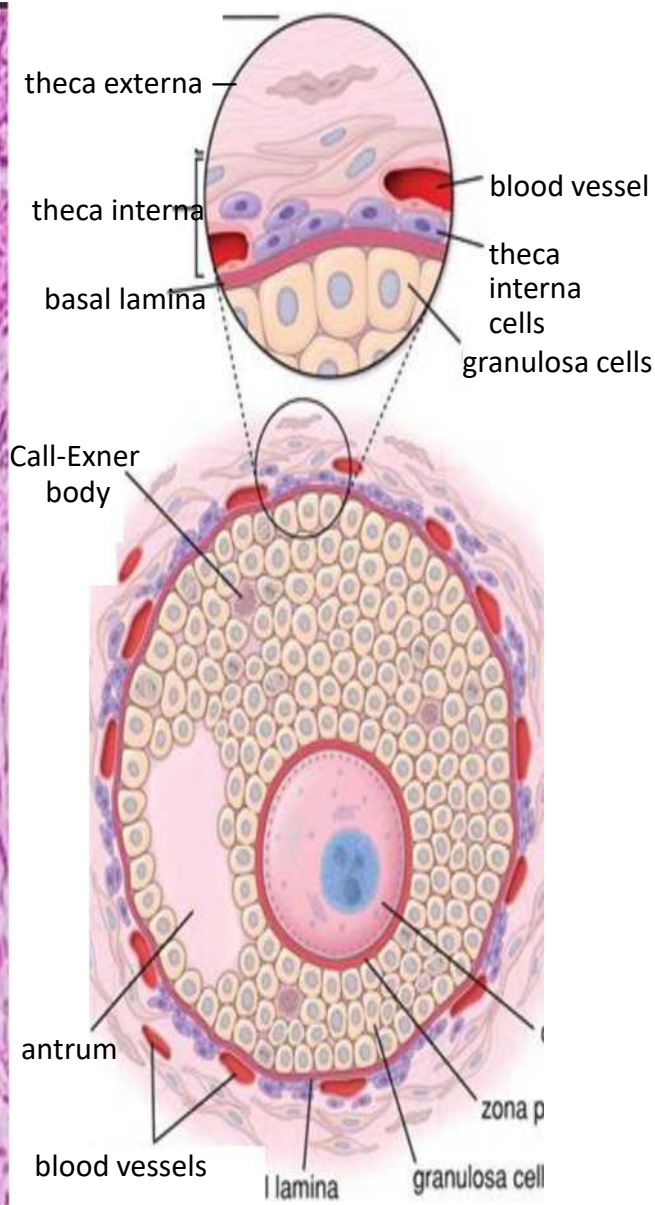
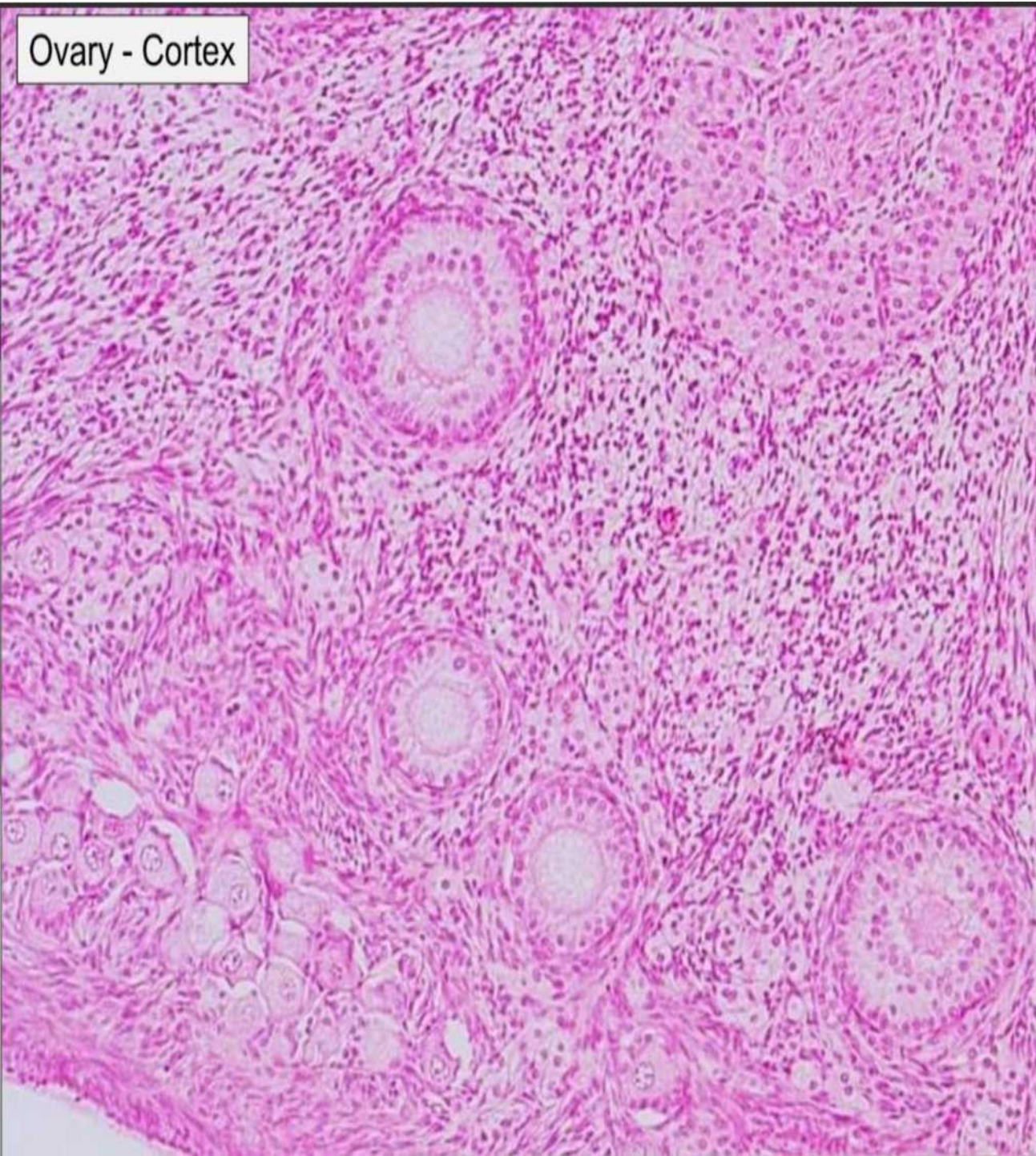
IV. For a definitive karyotypic diagnosis, either Chorionic Villus Sampling (CVS) or Amniocentesis should be performed. Describe how each test is performed, the gestational timing at which it is performed, and mention one advantage and one disadvantage of CVS in comparison to Amniocentesis.

7. As we noticed the initial vaginal bleeding in this patient, which blood vessels may be responsible for this bleed?

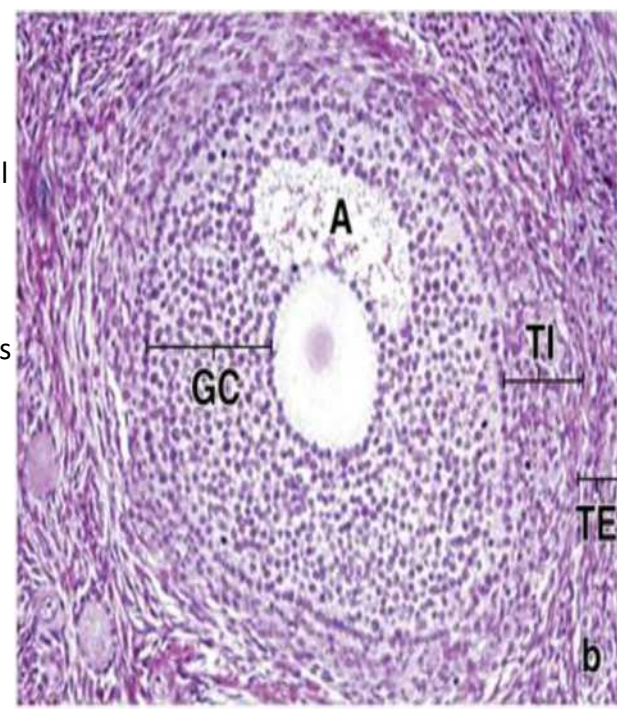
8 a. What are the changes occurring in the ovaries and uterus for the menstrual cycle to occur? Include in your discussion hormonal correlations with histological changes at the different phases (menstrual cycle, ovarian cycle and pregnancy).

8 b. What changes may occur if there is pregnancy?

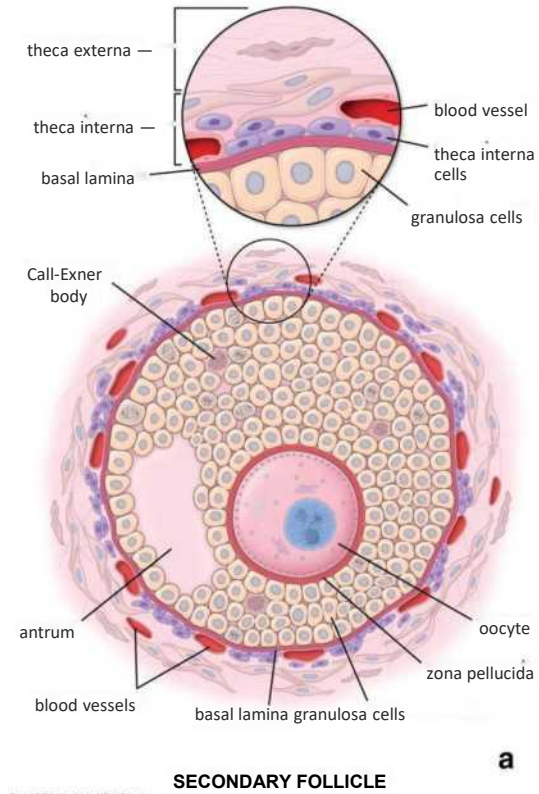
Ovary - Cortex



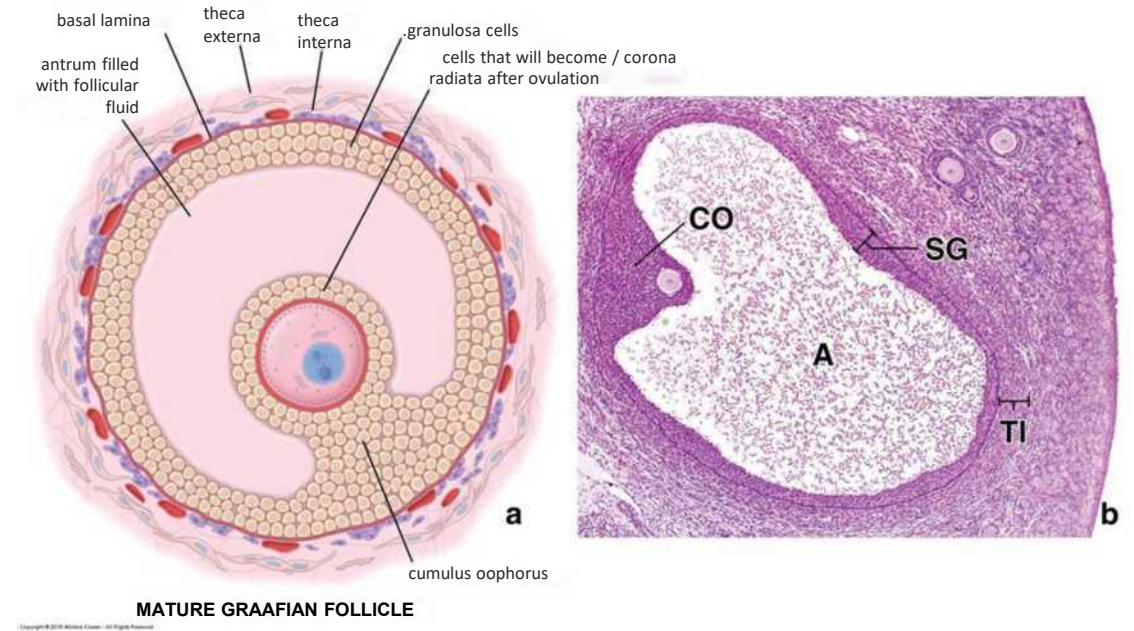
SECONDARY FOLLICLE **a**

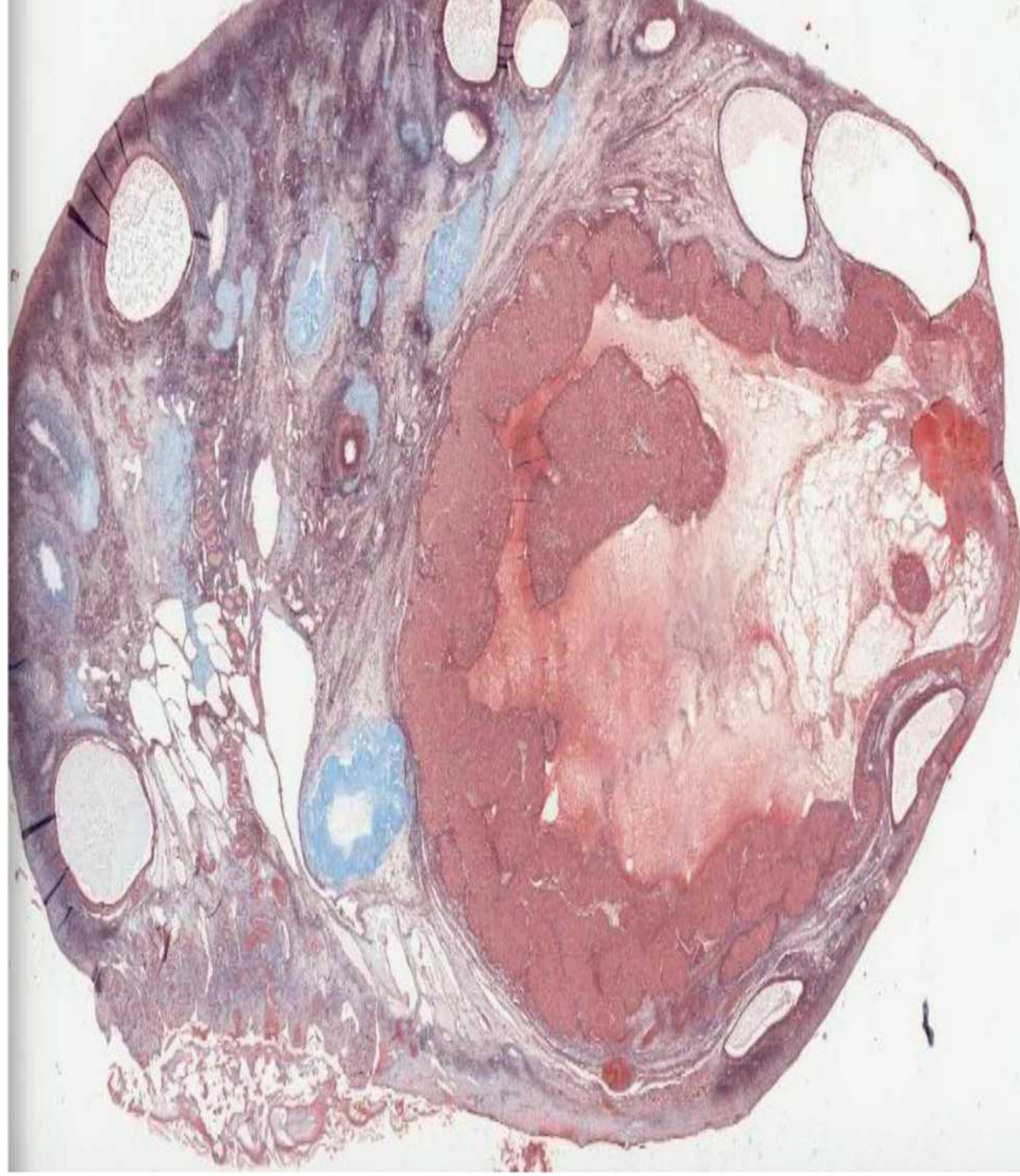


Ovary - Secondary Follicle

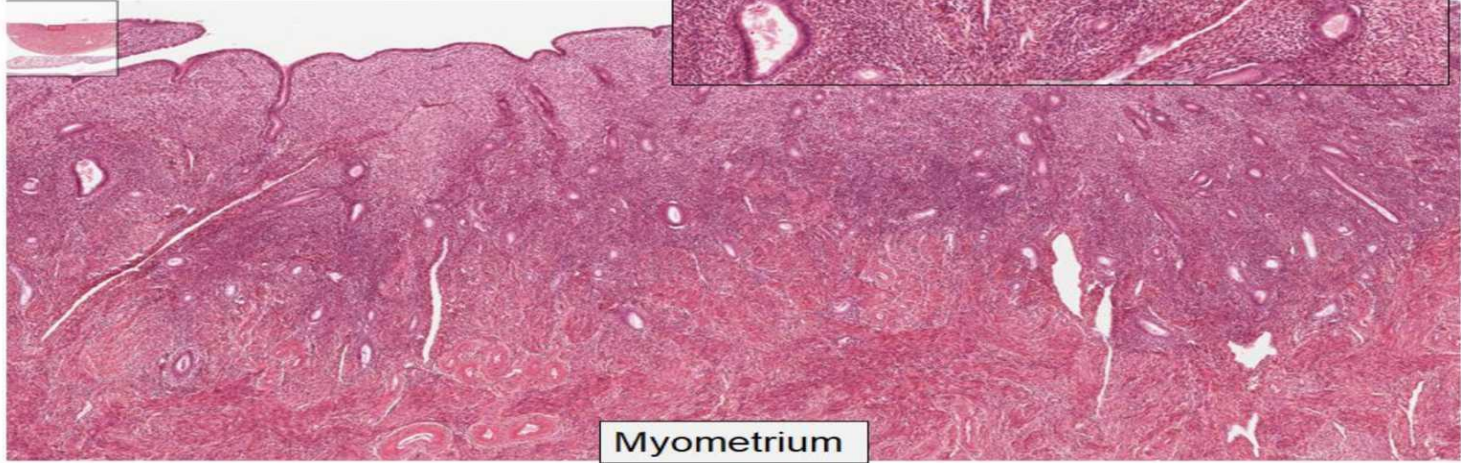


Ovary - Mature Graafian (tertiary) Follicle

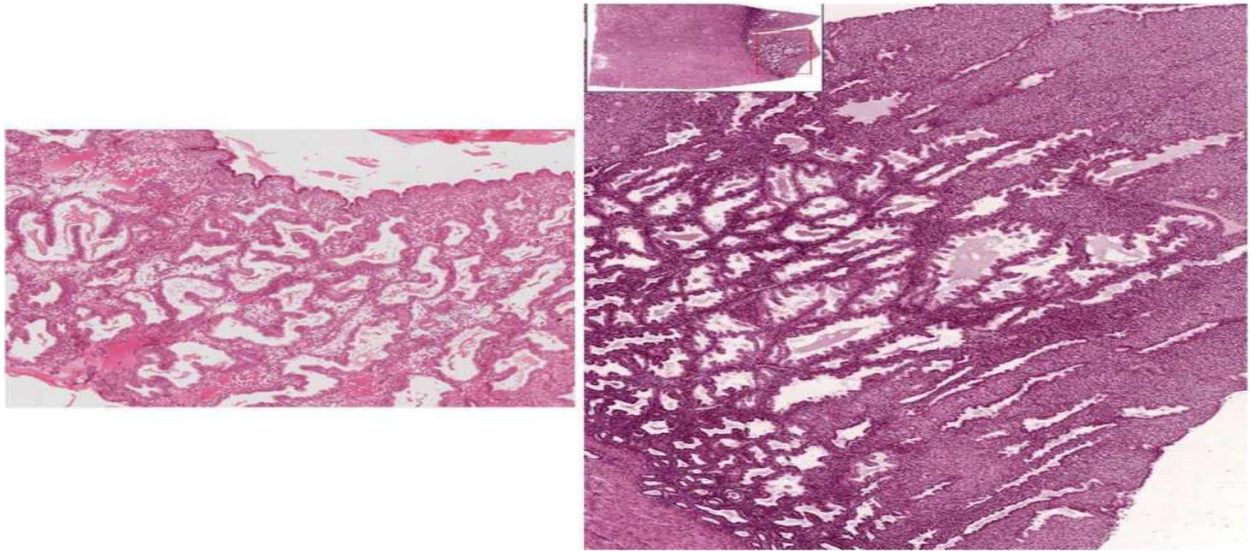




Endometrium - Proliferative Phase

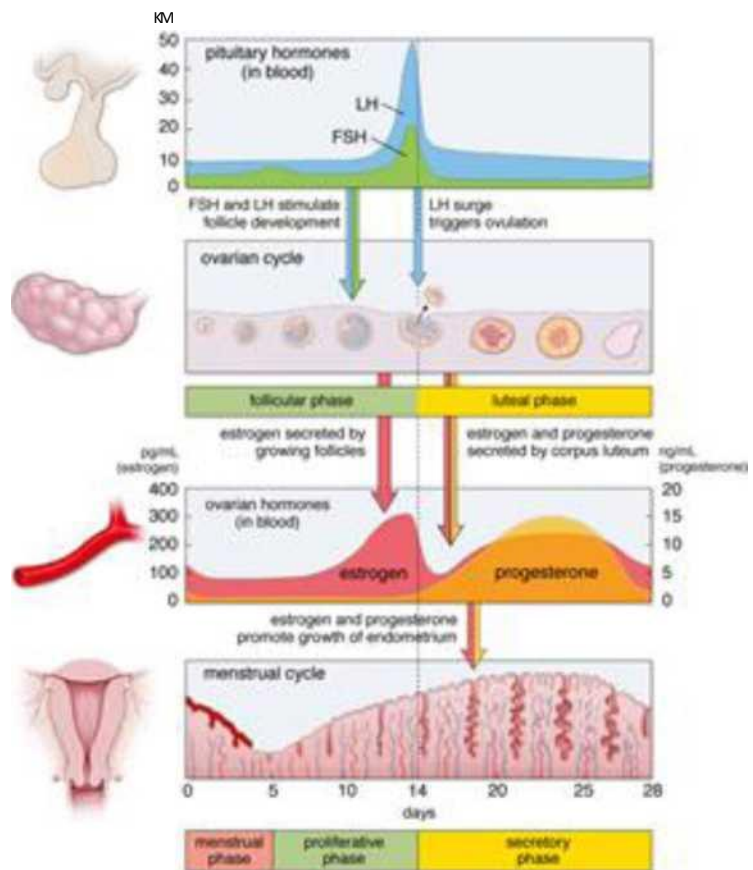


Endometrium - Late Secretory Phase



Early Secretory Phase





As she attained her full-term pregnancy, she delivered at 39th week. In regard the preparing of the mother for vaginal delivery a lateral episiotomy was performed.

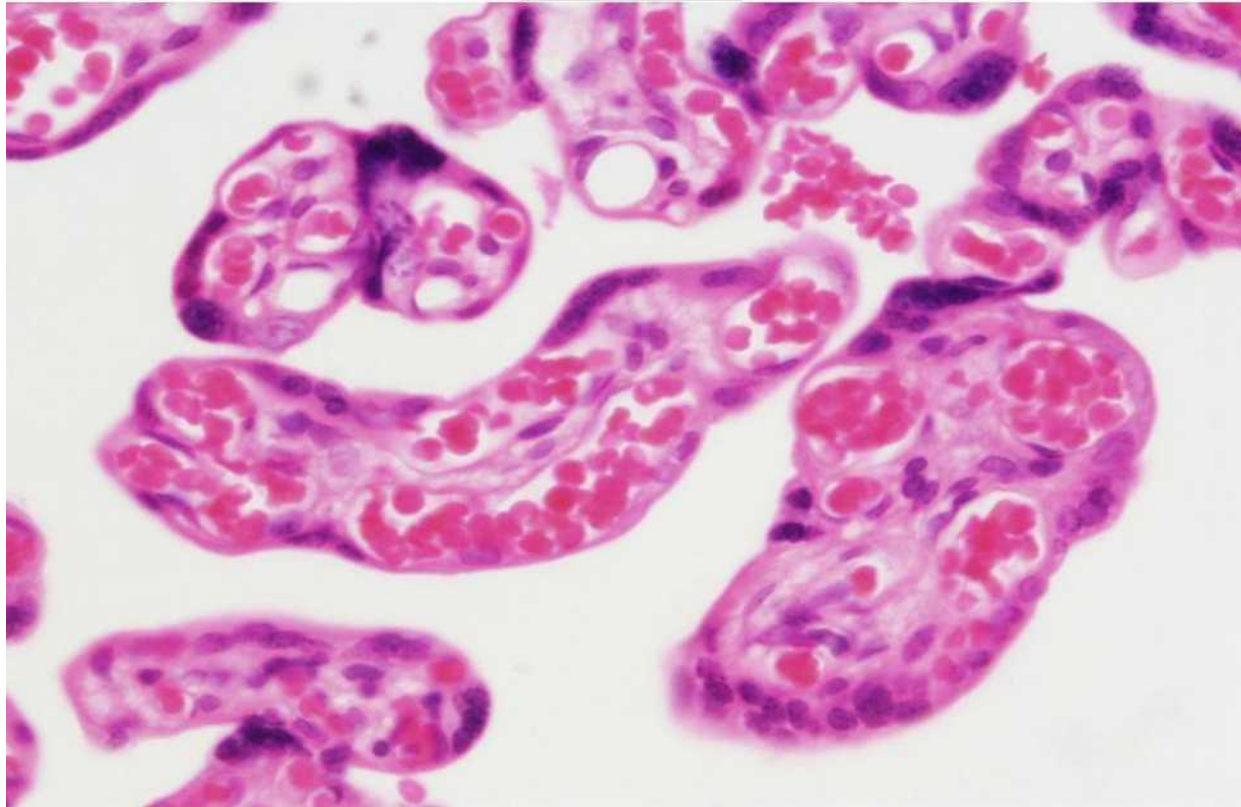
9. Discuss the innervation and lymphatic drainage of the perineum?

10. What important structures are avoided by doing a lateral episiotomy?

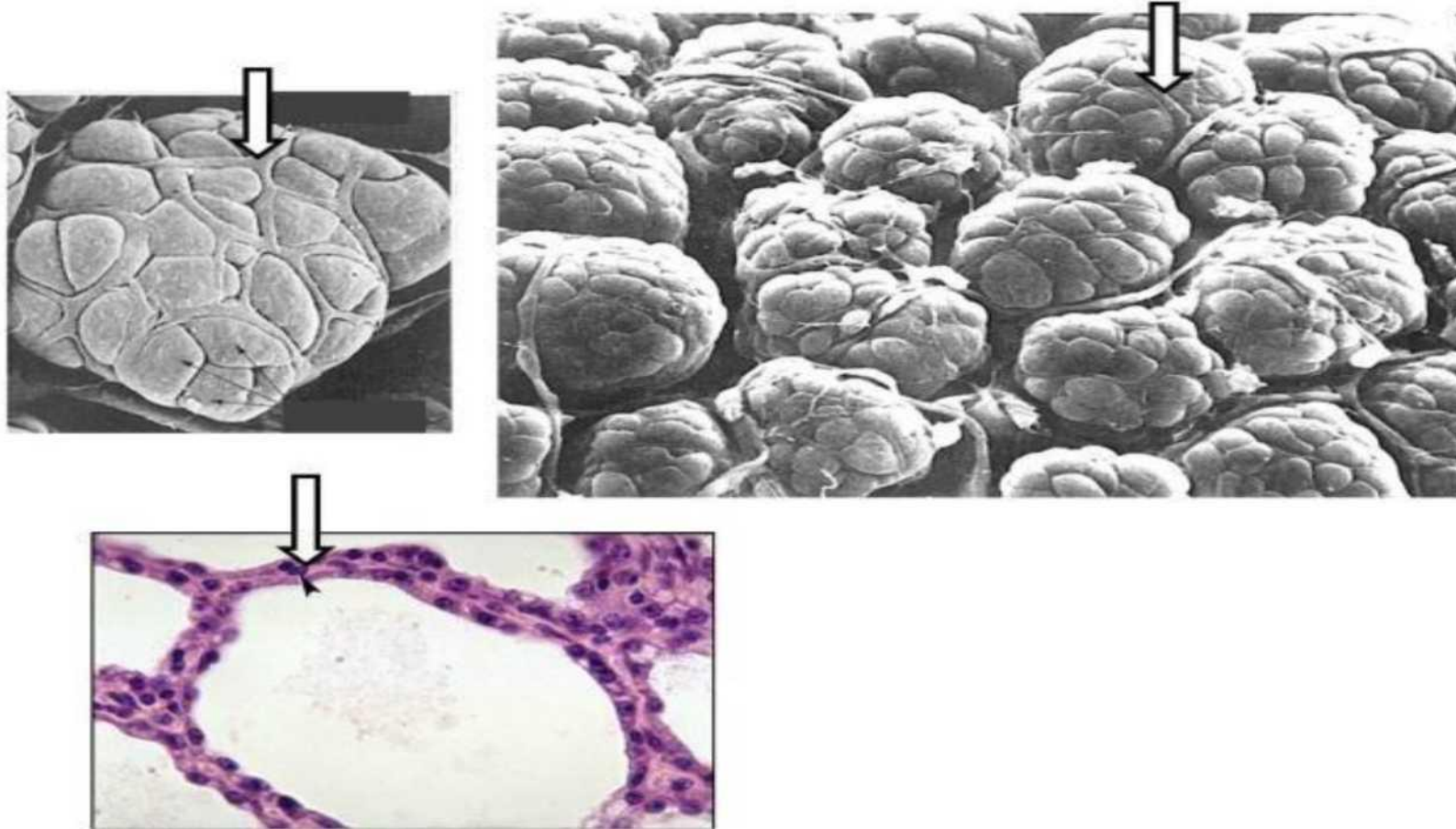
The placenta was delivered and found intact upon its examination.

11. What are the functions of placenta and discuss briefly?
12. Discuss briefly the different histological stages of placental development?

Placenta



13. Compare the histological features of a pubertal mammary gland and a lactating mammary gland. Include hormonal influences.



Late Pregnancy to Lactating Mammary Gland

